Question Paper - Report

27-Sep-2025 17:30:22
AKSHAT SRIVASTAVA .
Logout

Back

Question Paper

MANIPAL ACADEMY OF HIGHER EDUCATION

I SEM B.TECH. in CSE
MIDTERM EXAMINATION, SEPT. 2025
Exam Date & Time: 17-Sep-2025 (08:30 AM - 10:00 AM)

PROBLEM SOLVING USING COMPUTERS [CSE 1071]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Identify which of the following is valid C expression. (0.5)

int my_num = 100,000; int my_num = 100000; int my num = 1000; int $my_num = 10000;

2) In C programming language, the size of an array variable is declared by (0.5)

Program user Programmer Software Declared automatically

3) Select which of the following is NOT possible with any 2 operators in C. (0.5)

Different precedence, same associativity | Different precedence, different associativity | Same precedence, different associativity | Same precedence, same associativity

4) Analyze the C code below and select the correct output. (0.5)

```
#include <stdio.h>
int main() {
    int x = 0;
    if(x<5)
        printf("ONE");
    if(++x==1)
        printf("TWO");
    if(x!=0)
        printf("THREE");
    return 0;
}
```

ONE ONETWO TWOTHREE ONETWOTHREE

5) Identify the most appropriate output for the below C program. (0.5)

```c
#include <stdio.h>
int main() {
    int x = 0;
    switch (x) {
        default: printf("D");
        case 1: printf("A");
        case 2: printf("B"); break;
        case 3: printf("C");
    }
    return 0;
}
```

C DAB DC DABC

6) Select the most appropriate output for the below C program.

```c
#include <stdio.h>
int main() {
    int i = 1, j = 1;
    while (i <= 2) {
        while (j <= 2) {
            printf("%d %d ", i, j);
            j++;
        }
        i++;
    }
    return 0;
}
```

(0.5)

1 1 1 2 2 1 2 2 1 1 1 2 1 1 1 2 2 1 Infinite loop

7) Analyze the C code below and select the correct output. (0.5)

```c
#include <stdio.h>
int main() {
    int n = 3;
    do {
        printf("%d", n);
    } while (--n);
    return 0;
}
```

321 32 3210 21

8) Identify the output for the below C program. (0.5)

```c
#include <stdio.h>
int main() {
    int ans=0;
    for(int i=0;i<=10;i++){
        if(i%2==0) continue;
        ans+=i;
    }
    printf("%d",ans);
    return 0;
}
```

10 15 20 25

9) Analyse and fill in the blanks for completing the C code snippet to rotate an array by d elements. (0.5)

Example: for the input array 1 2 3 4 5
rotated array by 1 element is **2 3 4 5 1**
rotated array by 3 elements is **4 5 1 2 3**

```c
int arr[10], d, n, p = 1;
    while (_______) {
        int last = arr[0];
        for (int i = 0; _______ i++) {
            arr[i] = arr[i + 1];
        }
        _____________
        p++;
    }
```

p < = d , i < n - 1 , arr[n - 1] = last; p < d, i < n, arr[n] = last; p >=d, i >n , arr[n+1] = last p >d, i >=n , arr[n] = last

10) The given array is arr = {3,4,5,2,1}. The number of swaps in bubble sort and selection sort respectively are _____ (0.5)

5 and 4 3 and 7 7 and 3 2 and 5

DESCRIPTIVE

Answer all the questions.

11) A local library maintains a list of book ID numbers (unique integers) that are not initially sorted. The librarian wants the list to be arranged in ascending order. (4)

Write a C program that:

1. Takes as input a list of book IDs (unsorted).

2. Sorts the IDs using Bubble Sort.

3. Displays the sorted list.

12) Illustrate with an example how a typical C program development environment handles compilation and linking. (3)

13) Write an algorithm and construct a flowchart to check a given number is prime or not. (3)

14) Write a C program to compute the summation for the following series using *while* loops up to n terms. The variable n is a user input.

$$1 + 3\left(\frac{1}{9}\right) + 12\left(\frac{1}{9}\right)^2 + 48\left(\frac{1}{9}\right)^3 + \cdots.$$ (3)

15) Using the syntax of the conditional operator, briefly explain its working. Write the equivalent code for the code snippet below using the conditional operator.

```
if(year%4 == 0) {
if( year%100 == 0) {
if (year%400 == 0)
 printf("%d is a leap year",year);
else
 printf("%d is not a leap year",year);
}else printf("%d is a leap year",year);
}else printf("%d is not a leap year",year);
```

(3)

16) Write a C program to read Two, One-Dimensional Arrays of same data type (integer type) and merge them into another One-Dimensional Array of same type. Accept an integer and delete that element from the merged array. (3)

```
Sample I/O:
Enter the number of elements in the first array: 3
Enter 3 elements for the first array:
1       3       4
Enter the number of elements in the second array: 2
Enter 2 elements for the second array:
22      33
Merged array is:
1 3 4 22 33
Enter the element to delete: 22
Final Array after deletion: 1 3 4 33
```

17) Analyze the following code and evaluate the expressions. Show all the intermediate steps for the expression evaluation process to arrive at the final results.

```
#include<stdio.h>
void main()
{
int i=3,j=4,k=8;
int z=4*i%5+4*j+5%i*j-3;
int c = (i & j) * 2 + (i | k) - (j ^ k) + (k <= j) * 4;
printf("%d\t%d",z, c);
}
```

(2)

18) Write a C program that reads a player's skill level L from the user and prints an appropriate title based on the following conditions using an ***else-if ladder*** statement.

i. L = 5 ? "Grandmaster"
ii. L = 4 ? "Expert"
iii. L = 3 ? "Advanced"
iv. L = 2 ? "Intermediate"
v. L = 1 ? "Beginner"
vi. For any other value ? "Unrated"

(2)

19) Given an array of integers (duplicates allowed), write a C program to print all the elements that appear only once (unique elements).

(2)

© 2025 All rights reserved. epm IP: 45.112.145.45 epCloud 1.5

^ Top